

Title: Identifying Optimal Locations for New Public Parks in Hamilton City Using GIS

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1. Introduction

This report investigates the spatial distribution and accessibility of public parks within Hamilton, New Zealand, using geographic data sourced through the ArcGIS Pro Online Portal. Public parks are necessary factors of urban environments, presenting spaces for recreation, community interaction, and improved mental and physical health. Equitable access to these green spaces is a key concern in urban planning, as disparities in park availability can reflect and reinforce broader social inequalities. This study employs GIS-based spatial analysis to identify areas of the city that are underserved by parks, particularly in relation to population density and the built environment. The findings are intended to inform future planning decisions aimed at enhancing urban livability and promoting equitable access to green infrastructure across Hamilton.

2. Literature Review

Urban green spaces play a crucial role in improving the quality of life in cities. They offer opportunities for recreation and exercise, contribute to mental well-being, mitigate urban heat effects, support biodiversity, and foster community cohesion. As cities like Hamilton continue to urbanize, equitable access to these spaces is increasingly recognized as essential for sustainable urban development and public health. This literature review explores key studies that highlight the importance of green space planning and the value of Geographic Information Systems (GIS) in supporting equitable urban development.

Wolch et al. (2014) argue that urban green space is a critical determinant of environmental justice, health equity, and sustainable development. They outline the relationship between green space access and health, particularly for marginalized populations. The authors identify a phenomenon known as "green space inequity," where wealthier communities enjoy better access to parks and recreational areas, while lower-income and minority neighborhoods often remain underserved. This inequity is exacerbated as cities expand and densify, leaving limited opportunities for new park development in already developed areas. They advocate for integrating GIS-based spatial analysis into urban planning as a strategy to reduce these disparities and promote environmental justice.

Similarly, Kabisch and Haase (2014) emphasize the importance of spatial equity in green space planning in European cities. They observe that park distribution is often uneven and disproportionately favors affluent areas, a pattern mirrored in many global cities. Their research employs GIS-based spatial analysis to evaluate the distribution of green spaces across different demographic groups. They introduce the concept of "green justice," which refers to fair access to environmental amenities across all socio-economic sectors. Their findings underscore the need for planning policies that prioritize underserved neighborhoods to reduce inequality and improve urban resilience.

In the context of New Zealand, urban planning policies increasingly incorporate green infrastructure as a core element of city development. The Ministry for the Environment's "Our Land 2021" report highlights the importance of land use planning and the protection of natural resources in response to climate change and urbanization (Ministry for the Environment, 2021). The report stresses the need for equitable distribution of parks and recreational spaces, particularly in high-density residential zones. Green spaces are viewed not only as recreational assets but also as essential urban infrastructure that supports ecological health, stormwater management, and carbon sequestration.

Access to green spaces has direct implications for public health. In a widely cited study, Maas et al. (2006) found a strong correlation between the presence of green spaces and improved self-perceived health among urban residents. The study concluded that individuals living within walking distance of parks or nature reserves were more likely to report better physical and mental health. Notably, the positive effects of green space access were more pronounced in densely populated urban areas, where exposure to nature is otherwise limited. These findings provide a compelling rationale for prioritizing green space development in areas of high population density.

Beyond theoretical considerations, GIS has emerged as a powerful tool for analyzing green space accessibility and guiding strategic planning. According to Zhang et al. (2011), GIS methods such as buffer analysis, network analysis, and spatial overlays allow researchers and planners to identify areas that lack adequate access to parks. Their national study in the United States demonstrated how modeling spatial accessibility to parks using GIS could reveal geographic disparities that are not apparent through traditional statistical methods. Their approach enabled the identification of communities that fall outside the service range of existing parks, thereby informing decisions on where new parks should be established.

The integration of GIS in urban green space planning facilitates evidence-based decision-making by enabling the analysis of multiple spatial variables, such as population density, land use, infrastructure, and existing park locations. For instance, spatial overlays can be used to intersect underserved zones

with demographic data to highlight priority areas for intervention. Buffer analyses help define the effective service radius of existing parks, while erasure and clipping tools allow planners to exclude unsuitable sites, such as those already developed. Such capabilities are particularly relevant in fast-growing cities like Hamilton, where land use decisions must balance development needs with environmental and social sustainability.

In the specific case of Hamilton city, which is experiencing rapid urban growth, the challenge lies in ensuring that new development projects incorporate adequate green space infrastructure. Urban intensification policies, while beneficial for reducing sprawl and promoting sustainable transport, can reduce the availability of open spaces if not carefully managed. A GIS-based framework for green space planning can help anticipate and address these challenges by identifying suitable locations for future parks based on population density, existing infrastructure, and gaps in service coverage.

Overall, the literature strongly supports the integration of GIS tools in urban green space planning. From promoting public health to advancing environmental justice, the equitable distribution of green spaces is essential for building sustainable and inclusive cities. This project builds upon the findings of these studies to develop a GIS-based methodology for identifying optimal locations for new public parks in Hamilton city, focusing on underserved high-density areas and avoiding conflict with existing infrastructure.

3. Data Options and Issues

This analysis utilized several spatial datasets relevant to Hamilton City. When attempting to download some data, authorization was required. Therefore, the data was retrieved through the ArcGIS Pro online portal.

- **Population Data(2024):** Provided by Statistics New Zealand, this dataset included total population figures by census mesh block. The attribute table was updated to include calculated population density.
- **Reservoirs:** Sourced from the Hamilton City Council's open data portal, these polygons represented the spatial extent of current reservoirs, including parks. This layer was primarily used for reference and to ensure that future parks avoid utility infrastructure.
- **Building Footprints:** Provided by LINZ, these polygons were used to exclude built-up areas from consideration as suitable park zones.

- **Road Network:** Sourced from Hamilton's transport GIS database, road shapefiles were used for contextual mapping.

A key limitation was the absence of real-time land use zoning and land ownership data. Additionally, the population data was based on the most recent census and may not reflect current growth or demographic shifts.

4. Assessment of GIS Models and Tools

- **Population Density Calculation:** Using the Field Calculator, a new field (Pop_Density) was created by dividing the total population by the area (in km²). Symbology was then applied to support visual interpretation.
- **Buffer Tool:** A 300-meter Euclidean buffer was applied to existing parks to represent typical walking-distance accessibility.
- **Erase Tool:** This tool was used to remove park buffer zones from the city's extent, helping to identify underserved areas.
- **Select by Attribute and Export Features Tool:** High population density areas were selected using the "Select by Attribute" function. Population values above the median were classified as high density, and the selected features were exported to a new layer named *High_Population*.
- **Intersect Tool:** This was used to identify areas where high population density overlapped with underserved regions.
- **Building Footprint Erasure:** Final potential park zones were refined by erasing building footprints to ensure the availability of open land.
- **Summary Statistics Tool:** This was applied with SA2 as the case field and no statistics field selected to generate a summary table named Park_Count_by_SA2. This table counts the number of optimal park zones within each suburb.

These tools are commonly used in suitability and accessibility analyses. However, the use of Euclidean buffers is a simplification; network-based travel buffers would provide a more realistic measure of accessibility.

5. Method and Results

The methodology involved a multi-step GIS-based spatial analysis to identify optimal locations for new public parks in Hamilton City, prioritizing areas with high population density and limited access to existing parks. The following procedures were undertaken using ArcGIS Pro:

1. Importing Spatial Layers

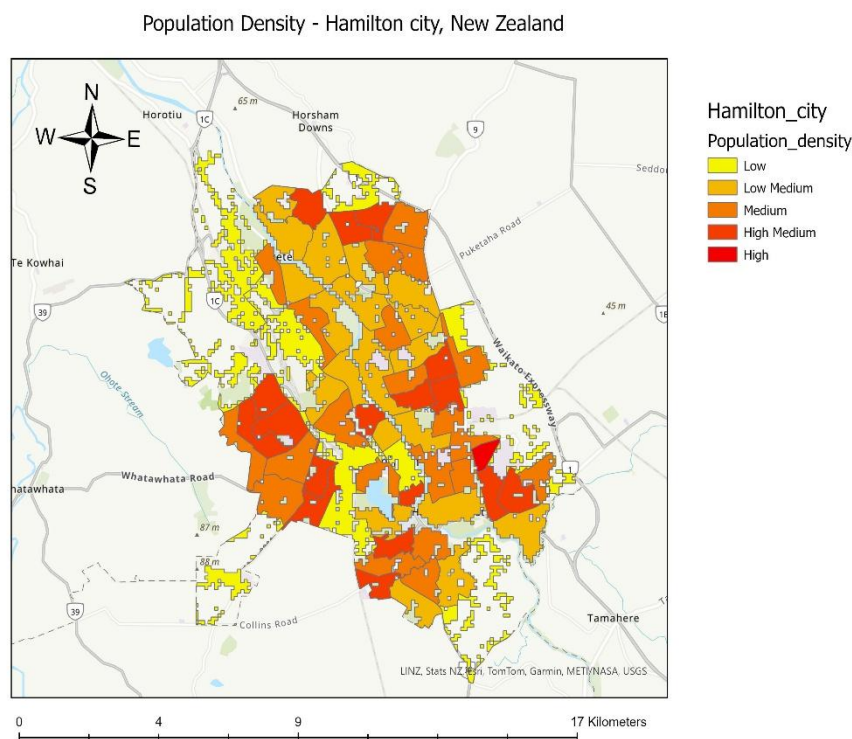
The initial step involved importing all relevant spatial datasets into the project environment. This included:

2. Population Density Calculation

A new field named Pop_Density was created in the population shapefile using the Field Calculator. Population density was calculated by dividing the total population by the polygon area in square kilometers. This step helped in identifying the spatial distribution of densely populated areas across the city.

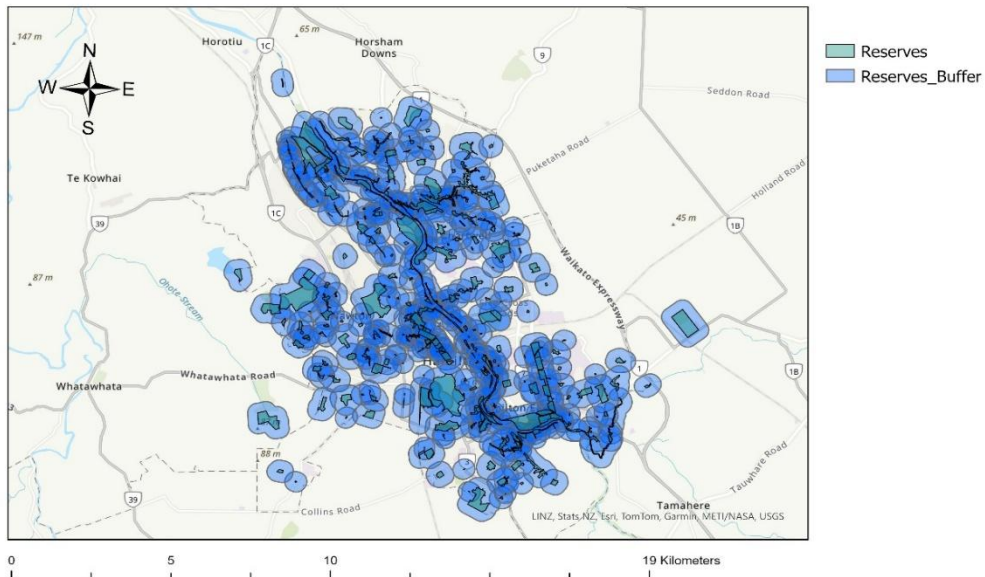
3. Applying Graduated Symbolology

Graduated color symbology was applied to the population layer based on the calculated density values. This visual differentiation allowed easy identification of high-density and low-density zones, forming the foundation for later spatial queries.



4. Creating a 300-Meter Buffer Around Existing Parks

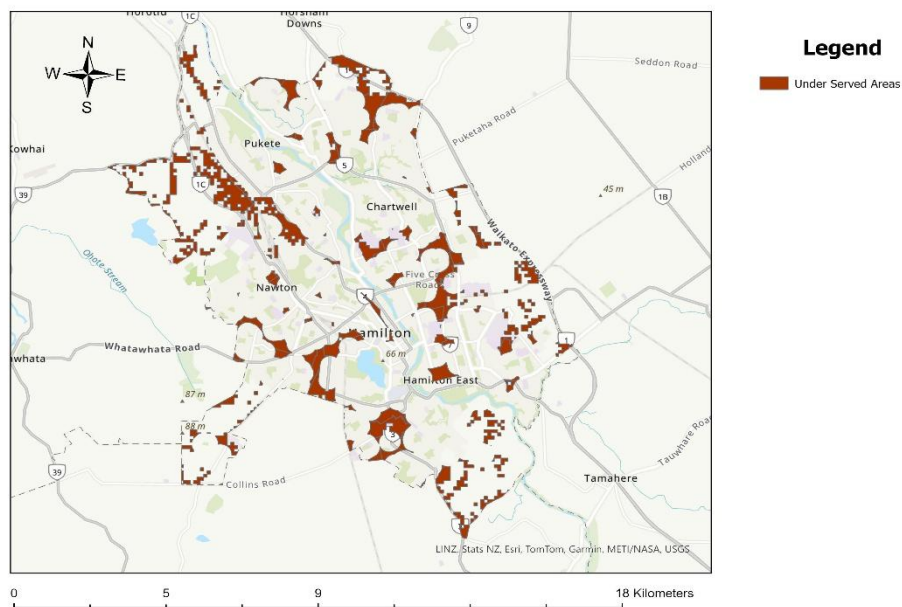
300-Meter Buffer Zone Around Reservoirs



A 300-meter Euclidean buffer was generated around the features representing parks within the reservoir layer. This buffer approximates the standard walking distance used in park accessibility studies and served to delineate zones already adequately serviced by green spaces.

5. Erase Tool to Identify Underserved Areas

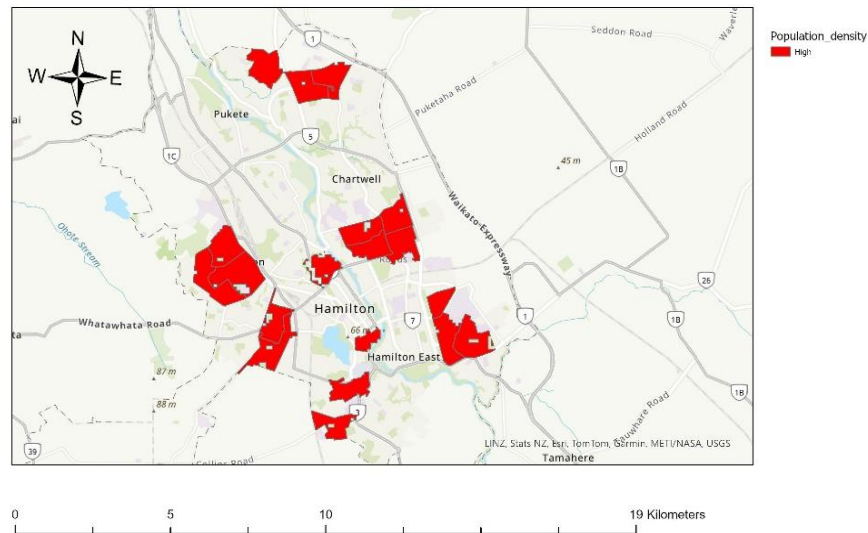
Underserved Regions Following Reservoir and Buffer Exclusion



The Erase tool was applied to subtract the buffered zones from the overall city extent. This operation highlighted areas that fall outside the 300-meter service radius designated as "underserved areas" lacking convenient access to existing parks.

6. Identifying High Population Density Areas

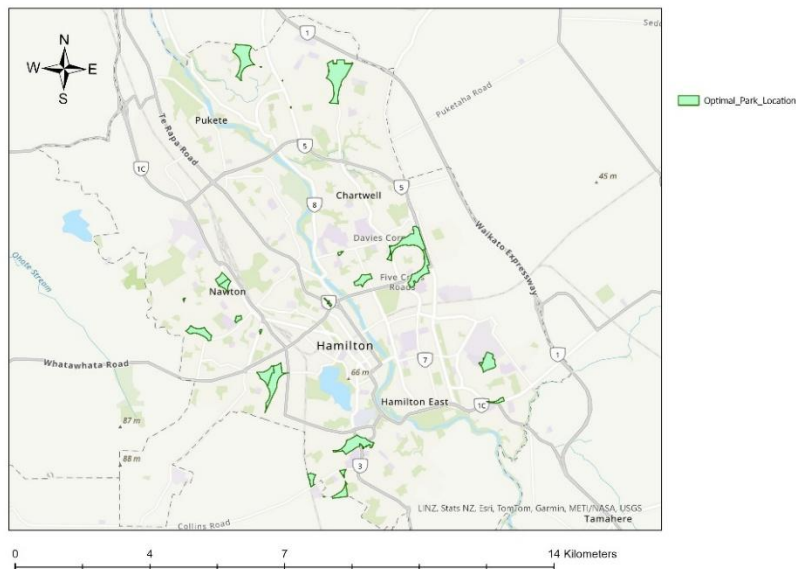
High Population Density Areas in Hamilton City



Using the “Select by Attribute” function, polygons with population densities above the median threshold were extracted. These features were exported into a new layer titled High_Population. These zones represent areas with the greatest potential need for new parks.

7. Intersecting High-Density and Underserved Areas

Identifying Optimal Areas for Establishing New Parks in Hamilton City



The Intersect tool was used to overlay the High_Population layer with the Underserved_Areas layer. This intersection identified spatial zones that are both densely populated and lack access to parks, making them high-priority candidates for intervention.

8. Excluding Developed Land Using the Erase Tool

To ensure that proposed park zones did not overlap with existing built structures, building footprint polygons were erased from the intersected result. This step refined the output to include only open, undeveloped parcels that are viable for new park development.

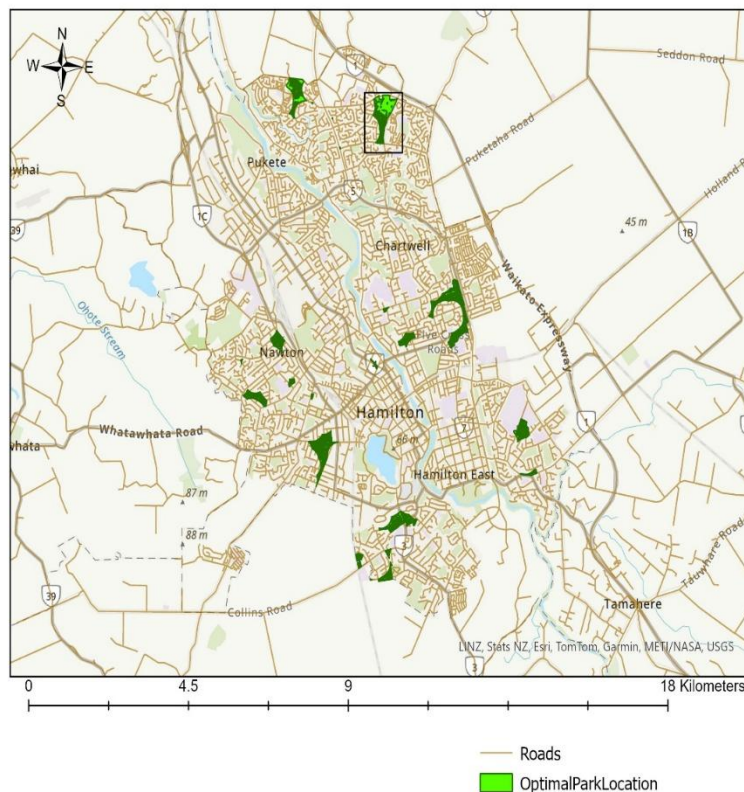
9. Final Output: Optimal Park Locations

The final result displayed spatially filtered areas that are:

- Outside the existing 300m park buffer
- Within high population density zones
- Free from building footprints

These areas were classified as optimal locations for establishing new parks.

Optimal Park Locations Excluding Existing Buildings



Zoomed-In View of Optimal Park Location Excluding



10. Suburb-Level Summary of Proposed Park Sites

To further analyze the geographic spread of optimal sites, the Summary Statistics tool was used. The optimal park layer was summarized by SA2 name (suburb), creating a table (Park_Count_by_SA2) that displays how many potential park zones exist per suburb. This provided insight into the distribution equity of the proposed locations.

Results:

The spatial analysis identified several underserved areas in Hamilton with high population density and insufficient access to existing parks. After excluding developed land and utility infrastructure, multiple open space parcels were identified as optimal locations for new parks.

To understand the distribution of these proposed park locations, a summary table was generated using the SA2 (suburb) name field. The table (Park_Count_by_SA2) shows how many optimal park zones were found in each suburb. Notably, **17 distinct suburbs** each had **one** proposed park location identified.

Optimal Park Locations

OBJECTID	SA2(Suburb)	Count
1	Crawshaw	1
2	Deanwell	1
3	Enderley North	1
4	Enderley South	1
5	Fairfield North (Hamilton City)	1
6	Fairfield South (Hamilton City)	1
7	Flagstaff North East	1
8	Hillcrest East (Hamilton City)	1
9	Hillcrest West (Hamilton City)	1
10	Kahikatea	1
11	Melville North	1
12	Nawton East	1
13	Nawton West	1
14	Rototuna East	1
15	Rototuna West	1
16	Swarbrick	1
17	Whitiora	1

This distribution indicates a relatively even spread of underserved areas across both central and suburban zones of Hamilton. The consistent count (1 optimal location per suburb) suggests that the spatial model performed equitably, identifying one key opportunity zone in each high-need area. These findings support the need for strategic park development in a broad range of neighborhoods rather than clustering them in a few areas.

6. Critical Evaluation

The GIS approach effectively highlighted gaps in park service coverage. However, some limitations affect the results:

- The use of Euclidean buffers ignores the actual street network, which may overestimate accessibility.
- Population data may not reflect current densities due to recent development.
- Land ownership and ecological constraints (e.g., slope, soil type) were not considered.
- Suitability was assessed only on physical proximity and population density, omitting social and economic criteria.

Despite these limitations, the analysis provides a strong foundation for more detailed urban planning studies.

7. Future Research

Further work should enhance the analysis through:

- Network-based service area analysis to model realistic walking distances.
- Integration of zoning, cadastral, and land ownership data to assess feasibility.
- Inclusion of demographic sub-groups (e.g., children, elderly) to target specific needs.
- Use of environmental layers (e.g., flood zones) to assess site viability.
- Community consultation to understand preferences and behavioral patterns.
- By combining GIS with social, environmental, and infrastructural data, more equitable and effective urban park planning can be achieved in Hamilton and similar cities.

References

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